Unit 6.01: Olympiad Circle Geometry, Power of a Point & Radical Axes

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 16x + 80 in the domain of Circle Geometry, Power of a Point & Radical Axes. By completing the square f_1x = x - h^2 + k , find the sum k + f_110 .

[ ] A: 44
[x] B: 36
[ ] C: 20
[ ] D: 34

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Complete the square: x^2 - 16x + 80 = x - 8^2 - 64 + 80 = x - 8^2 + 16 , giving minimum value k = 16 . **[M1]** 2. Evaluate at x = 10 : f_110 = 2^2 + 16 = 20 . **[M1]** 3. Sum the two quantities: k + f_110 = 16 + 20 = 36 . **[A1]** - **Final Verified Answer:** 36 .

Q2. A mathematical model in Circle Geometry, Power of a Point & Radical Axes is defined by the cubic function y_2x = 4x^3 - 8x^2 + 15x - 13 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 123
[ ] B: 60
[x] C: 75
[ ] D: 35

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 12x^2 - 16x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 129 - 163 + 15 = 75 . **[M1][A1]** - **Final Verified Answer:** 75 .

Q3. In Circle Geometry, Power of a Point & Radical Axes, two displacement vectors are given by mathbf{u}_{3} = langle 8, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 16
[ ] B: 128
[ ] C: 256
[x] D: 144

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 89 + -87 = 72 + -56 = 16 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 8^2 + -8^2 = 128 . **[M1]** 3. Combine: 16 + 128 = 144 . **[A1]** - **Final Verified Answer:** 144 .

Q4. The linear growth profile shown in the diagram for Circle Geometry, Power of a Point & Radical Axes follows an arithmetic progression with first term a_1 = 9 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[x] A: 765
[ ] B: 93
[ ] C: 855
[ ] D: 1395

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 9, d = 6 : S_{15} = frac{15}{2}[18 + 14 times 6] = frac{15}{2} times 102 = 765 . **[M1][A1]** - **Final Verified Answer:** 765 .

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Q5. In Circle Geometry, Power of a Point & Radical Axes, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 32x,mathrm{d}x .

[ ] A: 136
[x] B: 112
[ ] C: 144
[ ] D: 57

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Integrate term by term: int 18x^2 + 32x,mathrm{d}x = left[6x^3 + 16x^2right]_0^{2} . **[M1]** 2. Substitute the upper and lower limits: 62^3 + 162^2 - 0 = 48 + 64 = 112 . **[M1][A1]** - **Final Verified Answer:** 112 .

Q6. In Circle Geometry, Power of a Point & Radical Axes, consider the quadratic function f_6x = x^2 - 18x + 169 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_616 .

[ ] A: 234
[ ] B: 137
[x] C: 225
[ ] D: 218

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Complete the square: x^2 - 18x + 169 = x - 9^2 - 81 + 169 = x - 9^2 + 88 , giving minimum value k = 88 . **[M1]** 2. Evaluate at x = 16 : f_616 = 7^2 + 88 = 137 . **[M1]** 3. Sum the two quantities: k + f_616 = 88 + 137 = 225 . **[A1]** - **Final Verified Answer:** 225 .

Q7. The graph illustrates the cubic model y_7x = 4x^3 - 9x^2 + 33x - 20 studied in Circle Geometry, Power of a Point & Radical Axes. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 801
[ ] B: 624
[ ] C: 111
[x] D: 657

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 12x^2 - 18x + 33 . **[M1]** 2. Substitute x = 8 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=8} = 1264 - 188 + 33 = 657 . **[M1][A1]** - **Final Verified Answer:** 657 .

Q8. In Circle Geometry, Power of a Point & Radical Axes, two displacement vectors are given by mathbf{u}_{8} = langle 9, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[x] A: 167
[ ] B: 37
[ ] C: 130
[ ] D: 237

Solution: **Mark Scheme & Examiner Solution Circle Geometry, Power of a Point & Radical Axes:** 1. Compute the inner product: mathbf{u}_{8} cdot mathbf{v}_{8} = 98 + -75 = 72 + -35 = 37 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{8}|^2 = 9^2 + -7^2 = 130 . **[M1]** 3. Combine: 37 + 130 = 167 . **[A1]** - **Final Verified Answer:** 167 .